ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 1 style (theory). Section A: objective questions (30) · Section B: mechatronic and measurement systems, actuators (solenoids, stepper and servo motors), pneumatic and hydraulic calculations and components, gear and belt drives, lubrication and maintenance (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Mechatronics combines …

A mechanical, electronic, control and computer engineering. B only mechanics and hydraulics.
C chemistry and biology. D civil and mining engineering.

1.1.2 The part of a measurement system that changes a physical quantity into an electrical signal is the …

A display. B transducer. C amplifier. D recorder.

1.1.3 A stepper motor with a step angle of 1,8° needs … steps for one revolution.

A 100 B 360 C 200 D 180

1.1.4 A 5/2 directional control valve has …

A 5 positions and 2 ports. B 5 ports and 2 positions.
C 5 cylinders. D only 2 ports.

1.1.5 In pneumatics, an FRL unit is a …

A filter, regulator and lubricator. B flow, relief and limit unit.
C force, ratio and load unit. D fuse, relay and lamp unit.

1.1.6 Hydraulic systems transmit power by means of …

A air. B steam. C hydraulic oil. D a gas only.

1.1.7 A small driver gear that drives a large driven gear gives …

A higher speed and lower torque. B lower speed and higher torque.
C the same speed and torque. D only a change of direction.

1.1.8 The position of a hobby servo motor is set by …

A the AC mains frequency. B a mechanical cam.
C the supply voltage only. D the width of a control pulse.

1.1.9 Maintenance done at fixed intervals to prevent breakdowns is called …

A preventive maintenance. B breakdown maintenance.
C corrective maintenance. D emergency maintenance.

1.1.10 The smallest change in input that a sensor can detect is its …

A range. B span. C resolution. D drift.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 An automatic washing machine is an example of a mechatronic system.

1.2.2 Sensitivity is the ratio of the change in output to the change in input.

1.2.3 A single-acting cylinder is extended and retracted by air pressure.

1.2.4 Compressed air must be filtered and dried before it enters pneumatic equipment.

1.2.5 Pneumatic systems usually give larger forces than hydraulic systems of the same size.

1.2.6 A stepper motor needs an encoder to move a set number of steps.

1.2.7 An idler gear changes the gear ratio of a simple gear train.

1.2.8 A V-belt transmits power by friction on the sides of the belt.

1.2.9 Predictive maintenance uses measurements such as vibration and temperature to plan repairs.

1.2.10 A solenoid valve is operated by an electromagnet.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Device that produces movement from an energy source	A	torque
1.3.2	Cylinder that uses air pressure to move in both directions	B	flow control valve
1.3.3	Valve that lets air flow in one direction only	C	actuator
1.3.4	Valve used to set the speed of a cylinder	D	drift
1.3.5	Motor that turns to an angle set by a pulse width	E	servo motor
1.3.6	Teeth on the driven gear divided by teeth on the driver gear	F	bearing
1.3.7	Turning effect of a force	G	double-acting cylinder
1.3.8	Reduces friction between a turning shaft and its housing	H	flexible coupling
1.3.9	Slow change in a sensor's output over time for the same input	I	non-return valve

1.3.10	Joins two shafts and allows a little misalignment	J	gear ratio

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: MECHATRONIC AND MEASUREMENT SYSTEMS

2.1 Define the term mechatronics. (2)

2.2 Name and explain the FOUR basic elements of a mechatronic system, using a robot vacuum cleaner as an example. (4)

2.3 Give TWO other examples of mechatronic products used in South African homes or factories. (2)

2.4 A temperature transducer gives an output of 2 mV for every °C. Its range is 0 °C to 150 °C.

2.4.1 Calculate the output voltage at 85 °C. (2)

2.4.2 Calculate the output span of the transducer. (2)

2.5 Explain the terms range and resolution of a sensor. (2)

[14]

QUESTION 3: ACTUATORS

3.1 Explain how a solenoid actuator produces movement. (3)

3.2 A stepper motor has a step angle of 1,8°. Calculate:

3.2.1 The number of steps needed to turn the shaft 90° (2)

3.2.2 The speed of the shaft in r/min when the driver sends 400 pulses per second (3)

3.3 A hobby servo moves to 0° with a 1 ms pulse and to 180° with a 2 ms pulse. Explain how the pulse width controls the servo and state the angle for a 1,5 ms pulse. (3)

3.4 Give ONE advantage and ONE disadvantage of EACH of the following actuators: electric, pneumatic and hydraulic. (6)

3.5 Give ONE use of a linear actuator in a robot or automated machine. (1)

[18]

QUESTION 4: PNEUMATICS AND HYDRAULICS

4.1 A double-acting cylinder has a bore of 50 mm and a piston-rod diameter of 20 mm. It works at a pressure of 600 kPa. Calculate:

4.1.1 The force when the cylinder extends (3)

4.1.2 The force when the cylinder retracts (3)

4.1.3 Explain why the retracting force is smaller. (1)

4.2 Explain what the numbers in a 5/2 valve and a 3/2 valve mean, and state which type of cylinder EACH valve is normally used with. (4)

4.3 Name the THREE parts of an FRL unit and give the function of EACH. (3)

4.4 A hydraulic press has a small piston with an area of 2 cm² and a large piston with an area of 50 cm². A force of 100 N is applied to the small piston. Calculate the force produced by the large piston. (3)

4.5 Give ONE safety precaution when working with compressed air. (1)

[18]

QUESTION 5: MECHANICAL DRIVES AND MAINTENANCE

5.1 A driver gear with 20 teeth turns at 1 500 r/min and drives a gear with 60 teeth. The input torque is 4 N·m and the efficiency is 95%. Calculate:

5.1.1 The gear ratio (1)

5.1.2 The speed of the driven gear (2)

5.1.3 The output torque (3)

5.2 A driver pulley with a diameter of 100 mm turns at 1 440 r/min and drives a pulley with a diameter of 240 mm. Calculate:

5.2.1 The speed of the driven pulley (2)

5.2.2 The linear speed of the belt (3)

5.3 Give TWO differences between a chain drive and a belt drive. (4)

5.4 Name and explain THREE types of maintenance. (3)

5.5 Give TWO functions of a lubricant in a gearbox. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER